

第五章 基本应用二

§ 5.1 中心力场

1. 一般性质

球对称性, 角动量守恒: $[\hat{L}, \hat{H}] = 0$

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(r)$$

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

定态 Schrodinger 方程:

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial \psi}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial \psi}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} \right] + V\psi = E\psi$$

分离变量: $\psi(r, \theta, \phi) = R(r) Y(\theta, \phi)$

$$\left\{ \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial R}{\partial r}) \frac{1}{R} - \frac{2m}{\hbar^2} [V(r) - E] \right\} + \left[\frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial Y}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] \frac{1}{Y} = 0$$

$$\text{角部分: } \frac{\partial^2 Y}{\partial \theta^2} + \sin \theta \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial Y}{\partial \theta}) - c r^2 \sin^2 \theta Y = 0$$

其中 $c = l(l+1)$ 解为 $Y_{lm}(\theta, \phi)$

$$\text{径向部分: } \frac{\partial}{\partial r} (r^2 \frac{\partial R}{\partial r}) - \frac{2m r^2}{\hbar^2} [V(r) + \frac{l(l+1)\hbar^2}{2m r^2} - E] R = 0$$

$$\text{令 } u = r \cdot R$$

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \left(\frac{u}{r} \right) \right) = \frac{\partial}{\partial r} r^2 \frac{1}{r^2} \left[\frac{\partial u}{\partial r} \cdot r - u \right] = \frac{\partial^2 u}{\partial r^2} r + \frac{\partial u}{\partial r} - \frac{\partial u}{\partial r} = r \frac{\partial^2 u}{\partial r^2}$$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2 u}{\partial r^2} + [V(r) + \frac{l(l+1)\hbar^2}{2m r^2}] u = E u$$

特征能量与 m 无关, 是 $2l+1$ 重简并的.

2. 角动量

$$[\hat{H}, \hat{L}^2] = 0, [\hat{H}, \hat{L}_z] = 0 \quad ([\hat{L}^2, \hat{L}_z] = 0)$$

有征态可表为 $|E, l, m\rangle$

$$\langle \vec{r} | E, l, m \rangle = R_{El}(r) Y_{lm}(\theta, \phi)$$

满足球 Bessel 方程:

$$\frac{d^2 R_{El}(r)}{dr^2} + \frac{2}{r} \frac{dR_{El}}{dr} + \left(\frac{2mE}{\hbar^2} - \frac{l(l+1)}{r^2} \right) R_{El} = 0$$

$\quad \quad \quad = k^2$

通解为 $R_{El}(r) = A J_l(kr) + B n_l(kr)$

$$= A (-1)^l x^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\sin x}{x} + B (-1)^{l+1} x^l \left(\frac{1}{x} \frac{d}{dx} \right)^l \frac{\cos x}{x}$$

要求 $\lim_{r \rightarrow 0} R_{El}(r) = 0$, $\lim_{r \rightarrow 0} J_l(kr) = 0$, 则 $R_{El}(r) = A J_l(kr)$

$$\langle \vec{r} | E, l, m \rangle = A J_l(kr) Y_{lm}(\theta, \phi)$$

$$\text{归一化: } \langle E, l, m | E', l', m' \rangle = \delta(E - E') \delta_{ll'} \delta_{mm'}$$

$$= \int_V \psi_{E,l,m} \psi_{E',l',m'}^* r^2 \sin \theta dr d\theta d\phi$$

$$= \int_0^\infty R_{E,l}(r) R_{E',l'}(r) r^2 dr \int_0^\pi \int_0^{2\pi} Y_{lm}(\theta, \phi) Y_{l'm'}^*(\theta, \phi) \sin \theta d\theta d\phi$$

$$= \int_0^\infty R_{E,l}(r) R_{E',l'}(r) r^2 dr \cdot \delta_{ll'} \delta_{mm'}$$

$$\therefore \int_0^\infty R_{E,l}(r) R_{E',l'}(r) r^2 dr = \delta(E - E') \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$|A|^2 \int_0^\infty J_l(kr) J_{l'}(kr') r^2 dr = \left(\frac{\partial E}{\partial k} \right)^{-1} \delta(k - k') = \frac{m}{k\hbar^2} \delta(k - k')$$

$$\int_0^\infty J_l(kr) J_{l'}(kr') r^2 dr = \frac{\lambda}{2k^2} \delta(k - k')$$

$$\Rightarrow A = \sqrt{\frac{2k^2}{\lambda}} \quad (m = n = 1) \quad ?$$

$r \rightarrow \infty$: $J_l(kr)$ 渐近展开式

$$J_l(kr) \xrightarrow{r \rightarrow \infty} \frac{1}{kr} \sin(kr - \frac{l\pi}{2})$$

$$= \frac{1}{2ik} \left(e^{-i\frac{l\pi}{2}} \frac{e^{ikr}}{r} - e^{i\frac{l\pi}{2}} \frac{e^{-ikr}}{r} \right)$$

发散球面波

会聚球面波

* 平面波 $\leftrightarrow$ 球面波

将平面波在球面波基底下展开

$$|k\rangle = \sum_{l=0}^{\infty} \sum_{m=-l}^l C_{lm} |E, l, m\rangle$$

$$\langle r | k \rangle = \sum_{l,m} C_{lm} \langle r | E, l, m \rangle$$

$$\frac{e^{i\vec{k}\cdot\vec{r}}}{(2\pi)^{3/2}} = \sum_{l,m} C_{lm} j_l(kr) Y_{lm}(\theta, \phi)$$

$\vec{k}$ 沿 z 轴 $\rightarrow$ 与 m 无关, 只保留 $m=0$ 项

$$\frac{e^{ikr\cos\theta}}{(2\pi)^{3/2}} = \sum_l C_{l0} j_l(kr) Y_{l0}(\theta, \phi)$$

3. 原子轨道

设 r_1, r_2 分别为原子核、电子的坐标

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla_1^2 - \frac{\hbar^2}{2m} \nabla_2^2 + V(|\vec{r}_1 - \vec{r}_2|)$$

利用质心运动定律, 可转化为单体问题

$$\text{质心坐标 } \vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}, \quad \text{相对坐标 } \vec{r} = \vec{r}_1 - \vec{r}_2$$

$$\text{总质量 } m = m_1 + m_2, \quad \text{约化质量 } \mu = \frac{m_1 m_2}{m_1 + m_2}$$

以下记 $\nabla_r = \nabla$

$$\nabla_1 = \nabla + \frac{m_1}{m_1 + m_2} \nabla_R$$

$$r_1 = r_1(R, r), \quad \frac{\partial}{\partial r_1} = \frac{\partial}{\partial R} \frac{\partial R}{\partial r_1} + \frac{\partial}{\partial r} \frac{\partial r}{\partial r_1}$$

$$\nabla_1^2 = \nabla^2 + \left(\frac{m_1}{m_1 + m_2}\right)^2 \nabla_R^2 + \frac{m_1}{m_1 + m_2} (\nabla \cdot \nabla_R + \nabla_R \cdot \nabla)$$

$$\nabla_2 = -\nabla + \frac{m_2}{m_1 + m_2} \nabla_R$$

$$\nabla_2^2 = \nabla^2 + \left(\frac{m_2}{m_1 + m_2}\right)^2 \nabla_R^2 - \frac{m_2}{m_1 + m_2} (\nabla \cdot \nabla_R + \nabla_R \cdot \nabla)$$

$$\hookrightarrow \hat{H} = -\frac{\hbar^2}{2} \left(\frac{1}{m} \nabla^2 + \frac{1}{m} \nabla_R^2 \right) + V(r)$$

$$= -\frac{\hbar^2}{2m} \nabla_R^2 - \frac{\hbar^2}{2\mu} \nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$\text{分离变量: } \psi(R, r) = \phi(R) \psi(r)$$

$$\left\{ \begin{array}{l} -\frac{\hbar^2}{2m} \nabla_R^2 \phi(R) = E_c \phi(R) \\ -\frac{\hbar^2}{2\mu} \nabla^2 \psi(r) - \frac{Ze^2}{4\pi\epsilon_0 r} \psi(r) = (E_e - E_c) \psi(r) = E \psi(r) \end{array} \right.$$

$$-\frac{\hbar^2}{2\mu} \nabla^2 \psi(r) - \frac{Ze^2}{4\pi\epsilon_0 r} \psi(r) = (E_e - E_c) \psi(r) = E \psi(r)$$

对于相对运动部分, 束缚态可表为

$$\psi(\vec{r}) = R_{Zl}(r) Y_{lm}(\theta, \varphi)$$

$$\text{径向: } -\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left[r^2 \frac{dR_{Zl}(r)}{dr} \right] + \left[-\frac{ze^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right] R_{Zl} = E R_{Zl}$$

代数解法:

$$\text{令 } \hat{H}' R_{Zl}(r) = E R_{Zl}(r)$$

$$a = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2}$$

$$\text{其中 } \hat{H}' = \frac{\hat{p}_r^2}{2\mu} + V_{eff}(r), \quad \hat{p}_r = -i\hbar \left(\frac{d}{dr} + \frac{1}{r} \right), \quad V_{eff}(r) = -\frac{ze^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2}$$

推导:

$$\begin{aligned} & -\frac{1}{2\mu} \frac{\hbar^2}{r^2} \frac{d}{dr} \left[r^2 \frac{dR_{Zl}(r)}{dr} \right] + \left[-\frac{ze^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right] R_{Zl} = E R_{Zl} \\ & = -\frac{\hbar^2}{2\mu} \left[\frac{d^2 R_{Zl}(r)}{dr^2} + \frac{2}{r} \frac{dR_{Zl}(r)}{dr} \right] + \left[-\frac{ze^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right] R_{Zl} = E R_{Zl} \\ & = \frac{(-i\hbar)^2}{2\mu} \left[\frac{d^2 R_{Zl}(r)}{dr^2} + \frac{1}{r} \frac{dR_{Zl}(r)}{dr} + \frac{d}{dr} \left[\frac{1}{r} R_{Zl}(r) \right] + \frac{1}{r^2} R_{Zl}(r) \right] + \left[-\frac{ze^2}{4\pi\epsilon_0} \frac{1}{r} + \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right] R_{Zl} = E R_{Zl} \\ & = \left[\frac{1}{2\mu} (-i\hbar)^2 \left(\frac{d}{dr} + \frac{1}{r} \right)^2 + V_{eff}(r) \right] R_{Zl}(r) = E R_{Zl}(r) \end{aligned}$$

Verif

验证:

$$(1) [\hat{r}, \hat{p}_r] = i\hbar$$

$$\begin{aligned} \text{证: } [\hat{r}, \hat{p}_r] f(r) &= -i\hbar \left[\hat{r} \cdot \frac{d}{dr} + \frac{1}{r} \right] f(r) \\ &= -i\hbar \left\{ r \frac{df}{dr} + f(r) - \frac{d}{dr} [r f(r)] - f(r) \right\} \\ &= -i\hbar [-f(r)] = i\hbar f(r) \end{aligned}$$

$$(2) [\hat{p}_r, \frac{1}{r}] = \frac{i\hbar}{r^2}$$

$$\begin{aligned} \text{证: } [\hat{p}_r, \frac{1}{r}] f(r) &= -i\hbar \left[\frac{d}{dr} + \frac{1}{r}, \frac{1}{r} \right] f(r) \\ &= -i\hbar \left\{ \frac{d}{dr} \left(\frac{f(r)}{r} \right) + \frac{f(r)}{r^2} - \frac{1}{r} \frac{df(r)}{dr} - \frac{f(r)}{r^2} \right\} \\ &= -i\hbar \left\{ \frac{\frac{df(r)}{dr} r - f(r)}{r^2} + \frac{f(r)}{r^2} - \frac{1}{r} \frac{df(r)}{dr} - \frac{f(r)}{r^2} \right\} \\ &= \frac{i\hbar}{r^2} f(r) \end{aligned}$$

定义算符 $\hat{a}_l, \hat{a}_l^\dagger$

$$\begin{cases} \hat{a}_l = \frac{a}{\sqrt{2}\hbar} \left[i\hat{p}_r - \frac{(l+1)\hbar}{r} + \frac{2\hbar}{(l+1)a} \right] \\ \hat{a}_l^\dagger = \frac{a}{\sqrt{2}\hbar} \left[-i\hat{p}_r - \frac{(l+1)\hbar}{r} + \frac{2\hbar}{(l+1)a} \right] \end{cases}$$

$$(1) [\hat{a}_l, \hat{a}_l^\dagger] = \frac{(l+1)a^2}{r^2} = \frac{\mu a^2}{\hbar^2} (\hat{H}_{l+1} - \hat{H}_l)$$

$$\text{证: } [\hat{a}_l, \hat{a}_l^\dagger]$$

$$= \frac{a^2}{2\hbar^2} \left[i\hat{p}_r, -\frac{(l+1)\hbar}{r} + \frac{2\hbar}{(l+1)a} \right]$$

$$= \frac{a^2}{\hbar^2} \cdot i \cdot -(l+1)\hbar \cdot \frac{i\hbar}{r^2} = \frac{(l+1)a^2}{\hbar}$$

$$\hat{H}_{l+1} - \hat{H}_l = \frac{\hbar^2}{2\mu r^2} \left[(l+1)(l+2) - l(l+1) \right] = \frac{2}{\mu r^2} (l+1)\hbar^2$$

$$2. \hat{H}_l' = \frac{\hbar^2}{\mu a^2} \left[\hat{a}_l^\dagger \hat{a}_l - \frac{2}{2(l+1)^2} \right]$$

$$\text{证: } \hat{a}_l^\dagger \hat{a}_l = \frac{a^2}{2\hbar^2} \left\{ \left[\frac{2\hbar}{(l+1)a} - \frac{(l+1)\hbar}{r} \right]^2 + \hat{p}_r^2 \right\}$$

$$= \frac{a^2}{2\hbar^2} \left[\frac{2^2\hbar^2}{(l+1)^2 a^2} + \frac{(l+1)^2\hbar^2}{r^2} - \frac{2\hbar^2}{ar} + \hat{p}_r^2 \right]$$

$$\frac{\hbar^2}{\mu a^2} \left[\hat{a}_l^\dagger \hat{a}_l - \frac{2}{2(l+1)^2} \right] = \frac{\hbar^2}{\mu a^2} \cdot \frac{a^2}{2\hbar^2} \left[\frac{(l+1)^2\hbar^2}{r^2} - \frac{2\hbar^2}{ar} + \hat{p}_r^2 \right]$$

$$= \frac{\hat{p}_r^2}{2\mu} + \frac{1}{2\mu} \left[\frac{(l+1)^2\hbar^2}{r^2} - \frac{2\hbar^2}{ar} \right]$$

$$= \frac{\hat{p}_r^2}{2\mu} + \frac{(l+1)^2\hbar^2}{2\mu r^2} - \frac{2\hbar^2}{\mu ar} = \hat{H}_l'$$

$$3) [\hat{a}_l, \hat{H}_l'] = (\hat{H}_{l+1} - \hat{H}_l) \hat{a}_l$$

$$\text{证: } [\hat{a}_l, \hat{H}_l'] = \frac{\hbar^2}{\mu a^2} \left[\hat{a}_l, \hat{a}_l^\dagger \hat{a}_l - \frac{2}{2(l+1)^2} \right]$$

$$= \frac{\hbar^2}{\mu a^2} [\hat{a}_l, \hat{a}_l^\dagger] \hat{a}_l = \frac{\hbar^2}{\mu a^2} \frac{\mu a^2}{\hbar^2} (\hat{H}_{l+1} - \hat{H}_l) \hat{a}_l$$

$$= (\hat{H}_{l+1} - \hat{H}_l) \hat{a}_l$$

$$(4) \hat{a}_l \hat{H}_l' = \hat{H}_{l+1}' \hat{a}_l$$

由 3) 易证。

$$\hat{a}_l \hat{H}_l' R_{z,l} = \hat{H}_{l+1}' (\hat{a}_l R_{z,l}) = \hat{a}_l \in R_{z,l} = \hat{H}_l' (\hat{a}_l R_{z,l})$$

$R_{z,l+1} = \hat{a}_l R_{z,l}$ 也是能量 $\hat{H}_l'$ 的本征态, 本征值为 z

1. 存在 - 上限 $l_{\max}$

$$\hat{a}_{l_{\max}} R_{z, l_{\max}} = 0$$

$$\begin{aligned} \hat{H}_{l_{\max}} R_{z, l_{\max}} &= \frac{\hbar^2}{2\mu a^2} \left[\hat{a}_{l_{\max}}^+ \hat{a}_{l_{\max}} - \frac{z^2}{2(l+1)^2} \right] R_{z, l_{\max}} \\ &= -\frac{z^2 \hbar^2}{2\mu a^2 (l_{\max}+1)^2} R_{z, l_{\max}} = z R_{z, l_{\max}} \end{aligned}$$

记 $n = l_{\max} + 1$

$$E_n = -\frac{z^2 \hbar^2}{2\mu a^2 n^2} = -\frac{1}{4\pi\epsilon_0} \frac{ze^2}{2a} \frac{1}{n^2}$$

近似认为 $\mu \approx m_e$

$$a = a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}, \text{ Bohr 半径}$$

$$n = 1, 2, 3, \dots$$

给定 n , $l = 0, 1, \dots, n-1$

对于给定的 E_n , 简并度为 $d = \sum_{l=0}^{n-1} (2l+1) = n^2$

求波函数的具体形式:

$$\psi_{nlm}(r, \theta, \varphi) = R_{nl}(r) Y_{lm}(\theta, \varphi)$$

$$\text{给定 } n, \hat{a}_{n-1} R_{n, n-1}(r) = 0$$

$$\frac{a}{r^2} \left[\frac{\partial}{\partial r} - \frac{n}{r} + \frac{z}{na} \right] R_{n, n-1}(r) = 0$$

$$\text{解得 } R_{n, n-1}(r) = C_{n, n-1} r^{n-1} e^{-\frac{z}{na} r}$$

$$\hat{a}_+ R_{n, l} = R_{n, l+1}$$

$$\hat{a}_+^+ R_{n, l+1}(r) = \frac{z^2}{2} \left[\frac{1}{(l+1)^2} - \frac{1}{n^2} \right] R_{n, l}(r) \leftarrow \hat{a}_+^+ \hat{a}_+ R_{n, l} = \hat{a}_+ R_{n, l+1}$$

$$R_{n, l+1}(r) = \frac{2n^2 l^2}{z^2 (n+1)(n-1)} \hat{a}_{l-1} R_{n, l}(r)$$

$$\hat{a}_+ R_{n, l+1} = \dots ?$$

$$\Rightarrow R_{n, l}(r) = \left[\frac{z^2}{(na)^3} \frac{(n-l-1)!}{2n(n+1)!} \right] e^{-\frac{1}{2} \rho} \rho^l L_{n-l-1}^{2l+1}(\rho)$$

§ 3.2 自旋.

1. 自旋表示

$\vec{S} = (S_x, S_y, S_z)$ 选取 (S^2, S_z) 表象.

$$S = \frac{1}{2}, \quad S_z = \frac{1}{2}\hbar \text{ 或 } -\frac{1}{2}\hbar$$

表示为 $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$S_z = \frac{1}{2}\hbar \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad S_z |\uparrow\rangle = \frac{1}{2}\hbar |\uparrow\rangle, \quad S_z |\downarrow\rangle = -\frac{1}{2}\hbar |\downarrow\rangle$$

定义梯子算符

$$\hat{S}_{\pm} = \hat{S}_x \pm i\hat{S}_y$$

$$\hat{S}_+ |\uparrow\rangle = 0, \quad \hat{S}_+ |\downarrow\rangle = \hbar |\uparrow\rangle$$

$$\hat{S}_- |\uparrow\rangle = \hbar |\downarrow\rangle, \quad \hat{S}_- |\downarrow\rangle = 0$$

$$\therefore \hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \hat{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\hat{S}_x = \frac{1}{2}\hbar \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{S}_y = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Pauli 矩阵:

$$\vec{S} = (S_x, S_y, S_z)$$

$$= \frac{1}{2}\hbar \vec{\sigma} = \frac{1}{2}\hbar (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$$

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

对易关系: $[\hat{\sigma}_\alpha, \hat{\sigma}_\beta] = \epsilon_{\alpha\beta\gamma} 2i \hat{\sigma}_\gamma$

反对易关系: $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} = 2\delta_{\alpha\beta}$

例: 在 S_z 表象中, 求 $\vec{S} \cdot \vec{n}$ 的本征态, 其中 $\vec{n}$ 是任意单位矢量.

设 $\vec{n} = \sin\theta \cos\varphi \vec{x} + \sin\theta \sin\varphi \vec{y} + \cos\theta \vec{z}$

$$\vec{S} \cdot \vec{n} = \frac{1}{2}\hbar \left[\sin\theta \cos\varphi \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin\theta \sin\varphi \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \cos\theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right]$$

$$= \frac{1}{2}\hbar \begin{pmatrix} \cos\theta & e^{-i\varphi} \sin\theta \\ e^{i\varphi} \sin\theta & -\cos\theta \end{pmatrix}$$

$$\vec{S} \cdot \vec{n} |\psi\rangle = \lambda |\psi\rangle$$

解得 $\lambda_1 = \frac{\hbar}{2} \quad |1\rangle = \begin{pmatrix} e^{-\frac{i\varphi}{2}} \cos \frac{\theta}{2} \\ e^{\frac{i\varphi}{2}} \sin \frac{\theta}{2} \end{pmatrix}$

$$\lambda_2 = -\frac{\hbar}{2} \quad |2\rangle = \begin{pmatrix} -e^{-\frac{i\varphi}{2}} \sin \frac{\theta}{2} \\ e^{\frac{i\varphi}{2}} \cos \frac{\theta}{2} \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta \end{pmatrix}$$

$$\begin{vmatrix} \cos \theta - \lambda & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta - \lambda \end{vmatrix} = \lambda^2 - \cos^2 \theta - \sin^2 \theta = \lambda^2 - 1 = 0 \rightarrow \lambda = \pm 1$$

$$\lambda = 1 : \begin{pmatrix} \cos \theta - 1 & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta - 1 \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta - 1 & e^{-i\varphi} \sin \theta \\ 0 & \frac{1}{1 - \cos \theta} (\sin^2 \theta - 1 + \cos^2 \theta) \end{pmatrix}$$

$$\textcircled{2} + \textcircled{1} \cdot e^{i\varphi} \frac{\sin \theta}{1 - \cos \theta}$$

$$|1\rangle = \begin{pmatrix} e^{-i\varphi} \sin \theta \\ 1 - \cos \theta \end{pmatrix} = \begin{pmatrix} e^{-\frac{i\varphi}{2}} \frac{\sin \theta}{1 - \cos \theta} \\ e^{\frac{i\varphi}{2}} \end{pmatrix} = \begin{pmatrix} e^{-\frac{i\varphi}{2}} \cot \frac{\theta}{2} \\ e^{\frac{i\varphi}{2}} \end{pmatrix} = \begin{pmatrix} e^{-\frac{i\varphi}{2}} \cos \frac{\theta}{2} \\ e^{\frac{i\varphi}{2}} \sin \frac{\theta}{2} \end{pmatrix}$$

$$\lambda = -1 : \begin{pmatrix} \cos \theta + 1 & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta + 1 \end{pmatrix} \rightarrow \begin{pmatrix} \cos \theta + 1 & e^{-i\varphi} \sin \theta \\ 0 & 1 - \cos \theta - \frac{\sin^2 \theta}{1 + \cos \theta} \end{pmatrix}$$

$$|2\rangle = \begin{pmatrix} e^{-i\varphi} \sin \theta \\ 1 + \cos \theta \end{pmatrix} = \begin{pmatrix} 2e^{-\frac{i\varphi}{2}} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ 2e^{\frac{i\varphi}{2}} \cos^2 \frac{\theta}{2} \end{pmatrix} \rightarrow \begin{pmatrix} e^{-\frac{i\varphi}{2}} \sin \frac{\theta}{2} \\ e^{\frac{i\varphi}{2}} \cos \frac{\theta}{2} \end{pmatrix}$$

2. kramers 简并.

角动量 ~ 角动量.

$$\text{时间反演: } \hat{Q} |S, S_z\rangle = (-1)^{S_z} |S, -S_z\rangle = i^{2S_z} |S, -S_z\rangle$$

$$= \hat{Q}^2 |S, S_z\rangle = \hat{Q} (i^{2S_z} |S, -S_z\rangle)$$

$$= (-i)^{2S_z} (i)^{-2S_z} |S, S_z\rangle = (-1)^{2S_z} |S, S_z\rangle$$

$$= (-1)^{2S} |S, S_z\rangle \quad (-1)^{2S} = (-1)^{2S_z} \quad \forall S_z \text{ 成立}$$

$\Rightarrow$ 对于半整数角动量, $\hat{Q}^2 = -1$

$$[\hat{Q}, \hat{H}] = 0 \rightarrow \hat{H} \hat{Q} |\psi\rangle = \hat{Q} \hat{H} |\psi\rangle = E \hat{Q} |\psi\rangle$$

对于半整数角动量的系统, $\hat{Q}, \hat{H}$ 有共同本征态, 存在 kramers 简并

存在两个能量相同的本征态, 互为时间反演态.

证: 要证 $|\psi\rangle$ 与 $\hat{Q} |\psi\rangle$ 是两个不同状态, 只需证明 $|\psi\rangle, \hat{Q} |\psi\rangle$ 正交.

$$\langle \psi | \hat{Q} | \psi \rangle = \sum_{ij} \langle \psi | i \rangle \langle i | \hat{Q} | j \rangle \langle j | \psi \rangle$$

$$= \sum_{ij} \psi_i^* u_{ij} \hat{K} \psi_j = \sum_{ij} \psi_i^* u_{ij} \psi_j^* = \sum_j \psi_j^* (-u_{ji}) \psi_i^*$$

$$= \sum_j \psi_j^* (-u_{ji}) \hat{K} \psi_i = -\langle \psi | \hat{Q} | \psi \rangle = 0$$

角动量算符性质

$$\hat{Q} \hat{S} \hat{Q}^{-1} = -\hat{S}, \quad \hat{Q} = \hat{U} \hat{K}$$

在 S_z 表象下, $\hat{S}_x, \hat{S}_z$ 为实数矩阵, $\hat{S}_y$ 为虚数矩阵.

$\hat{K}$ 使 $\hat{S}_y$ 反号, 但不会影响 $\hat{S}_x, \hat{S}_z$

猜 $\hat{U} = e^{-\frac{i\hbar \hat{S}_y}{\hbar}}$, 绕 y 轴旋转 π .

$$\hat{Q} = e^{-\frac{i\hbar \hat{S}_y}{\hbar}} \hat{K}$$

$$= e^{-i\frac{\pi}{2} \hat{S}_y} \hat{K}$$

$$= (\cos \frac{\pi}{2} - i \sin \frac{\pi}{2}) \hat{S}_y \hat{K} = -i \hat{S}_y \hat{K}$$

$$\hat{Q}^2 = -1$$

$$\hat{Q} = -\hat{Q}^T$$

$$\hat{Q}^2 = \hat{U} \hat{U}^* = -I$$

$$\hat{U}^T \hat{U}^T = -I \quad \text{两式取转置}$$

$$\hat{U}^T = -\hat{U}$$

Stern-Gerlach 实验:

$$\vec{F} = -\nabla(-\vec{\mu} \cdot \vec{B}) = \vec{\mu} \cdot \nabla \vec{B}$$

例: 假设用 α 做 Stern-Gerlach 实验, 磁场强度为 $B_x = B_y = 0$, $B_z = B_0 + \lambda z$,

α 子质量为 m , 磁矩为 μ , 在磁场中的势能为 $H_\mu = -\mu \hat{S} \cdot \vec{B}$

在 $t=0$ 时刻, α 子的状态可表为 $|\psi_0\rangle = \psi(r, 0)(\alpha|\uparrow\rangle + \beta|\downarrow\rangle)$

且 $\langle r \rangle = 0$, $\langle p_x \rangle = p_0$, $\langle p_y \rangle = \langle p_z \rangle = 0$

计算经过时间 t 后, 自旋向上、下的 α 子 z 坐标期望值。

解: $\hat{H} = \hat{H}_p + \hat{H}_\mu = \frac{\hat{p}^2}{2m} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \mu(B_0 + \lambda \hat{S}_z) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$[\hat{H}, \hat{S}_z] = 0 \rightarrow$ 可以单独考虑粒子自旋向上、向下的运动。

Ehrenfest 定理:

$$\frac{d}{dt} \langle z_{\uparrow} \rangle = \frac{1}{m} \langle p_{z,\uparrow} \rangle, \quad \frac{d}{dt} \langle p_{z,\uparrow} \rangle = -\partial V(r) = \mu\lambda$$

$$\frac{d}{dt} \langle z_{\downarrow} \rangle = \frac{1}{m} \langle p_{z,\downarrow} \rangle, \quad \frac{d}{dt} \langle p_{z,\downarrow} \rangle = -\partial V(r) = -\mu\lambda$$

$$\hookrightarrow \langle z_{\uparrow} \rangle = \frac{\mu\lambda t^2}{2m}, \quad \langle z_{\downarrow} \rangle = -\frac{\mu\lambda t^2}{2m}$$

§ 3.3 角动量相加

$\vec{J} = \vec{J}_1 + \vec{J}_2$, 满足角动量的对易关系:

$$[\hat{J}_\alpha, \hat{J}_\beta] = \varepsilon_{\alpha\beta\gamma} i\hbar \hat{J}_\gamma$$

$$H = H_1 \otimes H_2$$

总角动量 $\vec{J} = \vec{J}_1 + \vec{J}_2$, 在 $(\vec{J}^2, J_z)$ 表象.

$$\vec{J}^2 |s, s_z\rangle = s(s+1) |s, s_z\rangle, \quad \hat{J}_z |s, s_z\rangle = s_z |s, s_z\rangle$$

选取其基态为直积态 $|s_1\rangle |s_2\rangle = |s_1, s_2\rangle$

记为 $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$

$$\hat{S}_z |\uparrow\uparrow\rangle = \hbar |\uparrow\uparrow\rangle, \quad \hat{S}_z |\uparrow\downarrow\rangle = 0 |\uparrow\downarrow\rangle$$

$$\hat{S}_z |\downarrow\downarrow\rangle = -\hbar |\downarrow\downarrow\rangle, \quad \hat{S}_z |\downarrow\uparrow\rangle = 0 |\downarrow\uparrow\rangle$$

$$\vec{S}^2 |\uparrow\uparrow\rangle = 2\hbar^2 |\uparrow\uparrow\rangle, \quad \vec{S}^2 |\downarrow\downarrow\rangle = 2\hbar^2 |\downarrow\downarrow\rangle$$

$$\vec{S}^2 |\uparrow\downarrow\rangle = \hbar^2 |\uparrow\downarrow\rangle + \hbar^2 |\downarrow\uparrow\rangle$$

$$\vec{S}^2 |\downarrow\uparrow\rangle = \hbar^2 |\downarrow\uparrow\rangle + \hbar^2 |\uparrow\downarrow\rangle$$

$$\vec{S}^2 = (\vec{S}_1 + \vec{S}_2)^2 = \vec{S}_1^2 + \vec{S}_2^2 + 2\vec{S}_1 \cdot \vec{S}_2 = \vec{S}_1^2 + \vec{S}_2^2 + 2(\hat{S}_{1x}\hat{S}_{2x} + \hat{S}_{1y}\hat{S}_{2y} + \hat{S}_{1z}\hat{S}_{2z})$$

$$= \vec{S}_1^2 + \vec{S}_2^2 + \frac{1}{2}(\hat{S}_{1+} + \hat{S}_{1-})(\hat{S}_{2+} + \hat{S}_{2-}) - \frac{1}{2}(\hat{S}_{1+} - \hat{S}_{1-})(\hat{S}_{2+} - \hat{S}_{2-})$$

$$= \vec{S}_1^2 + \vec{S}_2^2 + \hat{S}_{1+}\hat{S}_{2-} + \hat{S}_{1-}\hat{S}_{2+} + 2\hat{S}_{1z}\hat{S}_{2z}$$

在 $|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle$ 的子空间中, $\vec{S}^2 = \hbar^2 \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$

$$\lambda_1 = 2, \quad |\psi_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 0, \quad |\psi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\vec{S}^2$ 和 S_z 的共同本征态 $|s, s_z\rangle$ 可分为两组:

$$(1) \quad s = 0 \quad |00\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad \text{自旋单态, 交换反对称}$$

$$(2) \quad s = 1 \quad |11\rangle = |\uparrow\uparrow\rangle$$

$$|10\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad \text{自旋三重态, 交换对称}$$

$$|1-1\rangle = |\downarrow\downarrow\rangle$$

例: $|\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ $|\psi_2\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$ 求系统处在自旋单重态, 三重态的几率

$$|\psi\rangle = |\downarrow\rangle \otimes (\cos\theta |\uparrow\rangle + \sin\theta |\downarrow\rangle) = \begin{pmatrix} 0 \\ \cos\theta \\ \sin\theta \end{pmatrix}$$

$$= \cos\theta |\downarrow\uparrow\rangle + \sin\theta |\downarrow\downarrow\rangle$$

单重态: $P_1 = |\langle 0,0|\psi\rangle|^2 = \left| \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \cdot (\cos\theta |\downarrow\uparrow\rangle + \sin\theta |\downarrow\downarrow\rangle) \right|^2$

$$= \frac{1}{2} \cos^2\theta$$

三重态: $P_2 = |\langle 1,1|\psi\rangle|^2 + |\langle 1,0|\psi\rangle|^2 + |\langle 1,-1|\psi\rangle|^2$

$$= 0 + \frac{1}{2} \cos^2\theta + \sin^2\theta = \frac{1}{2} \cos^2\theta + \sin^2\theta$$

2. C-系数.

$$\vec{J} = \vec{J}_1 + \vec{J}_2$$

记其基矢为 $|j_1, m_1; j_2, m_2\rangle$

$$[\vec{J}^2, \vec{J}_1^2] = 0, [\vec{J}^2, \vec{J}_2^2] = 0$$

在 $\vec{J}_1^2, \vec{J}_2^2, \vec{J}_1^2, \vec{J}_2^2$ 的共同本征空间, $[\vec{J}_2, \vec{J}_1^2] = 0, [\vec{J}_2, \vec{J}_2^2] = 0$

$$|j, m; j_1, j_2\rangle$$

如何由 $|j_1, m_1; j_2, m_2\rangle$ 表示: 展开 $= \min(j_1, m_1 + j_2) \leq \text{C-系数}, m_1 + j_2 - m + 1$

$$|j, m; j_1, j_2\rangle = \sum |j_1, m_1; j_2, m_2\rangle \langle j_1, m_1; j_2, m_2 | j, m; j_1, j_2 \rangle \dots$$

给定 j : $m \in (-j, j)$, 有 $2j+1$ 种取值.

给定 m : $m = m_1 + m_2$ $m_1 \in [-j_1, j_1], m_2 \in [-j_2, j_2]$

(m_1, m_2) 有 $\min(j_1, m_1 + j_2) + \min(j_2, m_1 + j_2) - m + 1$ 种取值.

证: $m_1 = m - m_2, m_1 \in \min(j_1, m + j_2) - \max(-j_1, m - j_2) + 1$

$$\textcircled{1} m - j_2 \geq -j_1, \max(-j_1, m - j_2) = m - j_2$$

$$m_1 \in \min(j_1, m + j_2) + j_2 + m + 1 = \min(j_1, m + j_2) + \min(j_2, m + j_2) - m + 1$$

$$\textcircled{2} m - j_2 < -j_1, \max(-j_1, m - j_2) = -j_1$$

$$m_1 \in \min(j_1, m + j_2) + j_1 + 1 = \min(j_1, m + j_2) + \min(j_2, m + j_2) - m + 1$$

j 的取值范围.

$j \leq j_1 + j_2$. 设 j 最小值为 $j_{\min}$

$$\sum_{j=j_{\min}}^{j_1+j_2} (2j+1) = \underline{(2j_1+1)(2j_2+1)} \quad \text{直接是 Hilbert 空间维数.}$$

$$\hookrightarrow j_{\min} = |j_1 - j_2| \quad (j_{\min} \geq 0)$$

计算系数:

$$\hat{J}_{\pm} |j_1, m; j_1, j_2\rangle = (\hat{J}_{1\pm} + \hat{J}_{2\pm}) \sum |j_1, m_1; j_2, m_2\rangle \langle j_1, m_1; j_2, m_2 | j, m; j_1, j_2\rangle$$

$$= \sqrt{j(j+1) - m(m\pm 1)} |j, m\pm 1; j_1, j_2\rangle$$

$$= \sum |j_1, m_1; j_2, m_2\rangle \langle j_1, m_1; j_2, m_2 | j, m; j_1, j_2\rangle$$

$$= \sqrt{j_1(j_1+1) - m_1(m_1\pm 1)} |j_1, m_1\pm 1; j_2, m_2\rangle \langle j_1, m_1\pm 1; j_2, m_2 | j, m; j_1, j_2\rangle$$

$$+ \sqrt{j_2(j_2+1) - m_2(m_2\pm 1)} |j_1, m_1; j_2, m_2\pm 1\rangle \langle j_1, m_1; j_2, m_2\pm 1 | j, m; j_1, j_2\rangle$$

$$\hookrightarrow \sqrt{j(j+1) - m(m\pm 1)} \langle j_1, m_1; j_2, m_2 | j, m\pm 1; j_1, j_2\rangle$$

$$= \sqrt{j_1(j_1+1) - m_1(m_1\pm 1)} \langle j_1, m_1\pm 1; j_2, m_2 | j, m; j_1, j_2\rangle$$

$$+ \sqrt{j_2(j_2+1) - m_2(m_2\pm 1)} \langle j_1, m_1; j_2, m_2\pm 1 | j, m; j_1, j_2\rangle$$

逆推:

$$J = j_1 + j_2$$

$$\text{对 } m = j_1 + j_2 : \langle j_1, j_1; j_2, j_2 | j_1 + j_2, j_1 + j_2; j_1, j_2\rangle = 1$$

$$\text{对 } m = j_1 + j_2 - 1$$

$$\sqrt{(j_1+j_2)(j_1+j_2+1) - (j_1+j_2)(j_1+j_2-1)} \langle j_1, j_1-1; j_2, j_2 | j_1 + j_2, j_1+j_2-1, j_1, j_2\rangle$$

$$= \sqrt{j_1(j_1+1) - (j_1-1)j_1} \langle j_1, j_1-1; j_2, j_2 | j_1+j_2, j_1+j_2-1, j_1, j_2\rangle$$

$$\hookrightarrow \langle j_1, j_1-1; j_2, j_2 | j_1 + j_2, j_1+j_2-1, j_1, j_2\rangle = \sqrt{\frac{j_1}{j_1+j_2}}$$

$$\text{同理: } \langle j_1, j_1; j_2, j_2-1 | j_1 + j_2, j_1+j_2-1, j_1, j_2\rangle = \sqrt{\frac{j_2}{j_1+j_2}}$$

3. 角动量耦合

$$H = H_0 + H_{\text{int}} = H_0 + \lambda \vec{S} \cdot \vec{L}$$

H_0 有本征态可表为轨道态与自旋态的直积 $|n, l, m\rangle |s, m_s\rangle$

$$\vec{J} = \vec{L} + \vec{S}, [\vec{J}, H] = 0$$

$$[\vec{H}, \vec{S}^2] = 0, [\vec{H}, \vec{L}^2] = 0$$

2(2l+1) 的简并分裂为 $J = l + \frac{1}{2}$ 、 $J = l - \frac{1}{2}$ 两组。

$$\vec{H} = H_0 + \frac{\lambda}{2} (\vec{J}^2 - \vec{S}^2 - \vec{L}^2)$$

简并度分别为 2l+2、2l

($S = \frac{1}{2}, -\frac{1}{2}$)

$$E = E_0 + \frac{\lambda}{2} [J(J+1) - S(S+1) - L(L+1)]$$

§ 3.4 带电粒子在电磁场中的运动

$$H = \frac{1}{2m} (\vec{p} - q\vec{A})^2 + q\phi$$

正则量子化.

$$\hat{H} = \frac{1}{2m} (\hat{\vec{p}} - q\vec{A})^2 + q\hat{\phi}$$

规范变换 $\left\{ \begin{array}{l} \vec{A} \rightarrow \vec{A}' = \vec{A} + \nabla \chi(\vec{r}, t) \\ \phi \rightarrow \phi' = \phi - \frac{\partial \chi(\vec{r}, t)}{\partial t} \end{array} \right.$

u(1)

设 $|\psi\rangle \rightarrow |\psi'\rangle = \hat{G}|\psi\rangle$, $\langle\psi'|\psi'\rangle = \langle\psi|\psi\rangle$

u $\hat{G} = e^{i\theta}$

动量算符: $\hat{\vec{p}} = \vec{p} - q\vec{A}$

动量期待值: $\vec{p} = \langle\psi|\hat{\vec{p}}|\psi\rangle$

$= \vec{p}' = \langle\psi'|\hat{\vec{p}}|\psi'\rangle$ (正则动量不变)

u $\hat{G}(\vec{p} - q\vec{A})\hat{G}^\dagger = \vec{p} - q\vec{A}'$ $|\psi'\rangle = \hat{G}|\psi\rangle$

$q(\vec{A}'\hat{G} - \hat{G}\vec{A}') = q\{\nabla\chi(\vec{r}, t) + [\vec{A}, \hat{G}]\}$

$= \hat{\vec{p}}\hat{G} - \hat{G}\hat{\vec{p}} = i[\hat{\vec{p}}, \hat{G}]$

$[\vec{A}, \hat{G}] = 0$,

$i[\hat{\vec{p}}, \hat{G}]\vec{f} = -i\hbar[\nabla(\hat{G}\vec{f}) - \hat{G}\nabla\vec{f}] = -i\hbar\nabla\hat{G}\cdot\vec{f}$

$\Rightarrow q\nabla\chi(\vec{r}, t) = -i\hbar\nabla\hat{G} = \hbar\nabla\theta(\vec{r}, t)$,

解得 $\theta(\vec{r}, t) = \frac{q\chi(\vec{r}, t)}{\hbar}$

$\hat{G} = e^{i\frac{q\chi(\vec{r}, t)}{\hbar}}$

规范变换下, schrodinger 方程不变.

证: 考虑规范变换

$$\begin{cases} \hat{A} \rightarrow \hat{A}' = \hat{A} + \vec{\nabla} \chi(\vec{r}, t) \\ \hat{\phi} \rightarrow \hat{\phi}' = \hat{\phi} - \frac{\partial \chi(\vec{r}, t)}{\partial t} \\ |\psi\rangle \rightarrow |\psi'\rangle = e^{i q \chi(\vec{r}, t) / \hbar} |\psi\rangle \end{cases}$$

$$\begin{aligned} & \left[\frac{1}{2m} (\vec{p} - q \vec{A})^2 + q \hat{\phi} \right] |\psi'\rangle \\ &= \frac{1}{2m} \vec{p}^2 - q \vec{p} \cdot \vec{A} + \frac{\partial \chi(\vec{r}, t)}{\partial t} - q \left[\hat{A} + \frac{\partial \chi(\vec{r}, t)}{\partial t} \right] \vec{p} + q^2 \vec{A}^2 + \end{aligned}$$

规范变换前后:

$$\begin{aligned} & \frac{1}{2m} [(\vec{p} - q \vec{A})^2 + q \hat{\phi}] |\psi\rangle = i \hbar \frac{\partial}{\partial t} |\psi\rangle \\ & \frac{1}{2m} [(\vec{p} - q \vec{A}')^2 + q \hat{\phi}'] |\psi'\rangle = i \hbar \frac{\partial}{\partial t} |\psi'\rangle \end{aligned}$$

连续性方程: $\frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \cdot \vec{J}(\vec{r}, t) = 0$, $\rho = \psi^* \psi$ 抗磁流密度.

$$\hookrightarrow \vec{J}(\vec{r}, t) = -\frac{i \hbar}{2m} \left[\psi^*(\vec{r}, t) \vec{\nabla} \psi(\vec{r}, t) - \psi(\vec{r}, t) \vec{\nabla} \psi^*(\vec{r}, t) \right] - \frac{q}{m} \vec{A} \psi^*(\vec{r}, t) \psi(\vec{r}, t)$$

$$\text{设 } \psi(\vec{r}, t) = |\rho(\vec{r}, t)| e^{i \theta(\vec{r}, t)}$$

$$\hookrightarrow \vec{J}(\vec{r}, t) = \frac{\rho(\vec{r}, t)}{m} \left[\hbar \vec{\nabla} \theta(\vec{r}, t) - q \vec{A} \right]$$

Coulomb 规范 $\vec{\nabla} \cdot \vec{A} = 0$ $(\vec{p}, \vec{A}) = -i \hbar \vec{\nabla} \cdot \vec{A} = 0$

$$\hat{H} = \frac{\hat{p}^2}{2m} - \frac{q}{m} \vec{A} \cdot \vec{p} + \frac{q^2}{2m} \vec{A}^2 + q \hat{\phi}$$

2. Zeeman 效应

原子处在均匀磁场 $\vec{B}$ 中

$$\hat{H} = \frac{1}{2m_e} (\vec{p} + e\vec{A})^2 - e\phi + U(\vec{r})$$

选取矢势 $\vec{A} = \frac{1}{2} \vec{B} \times \vec{r}$, $\phi = 0$

$$\begin{aligned} \vec{v} \times \vec{A} &= \frac{1}{2} \left[\partial_y (B_x y - B_y x) - \partial_z (B_z x - B_x z) \right] \hat{z} \\ &= \frac{1}{2} (2B_x \hat{y} - 2B_y \hat{x}) = \vec{B} \end{aligned}$$

$$\vec{v} \cdot \vec{A} = \frac{1}{2} (\vec{v} \cdot \vec{B}) \times \vec{r} = 0$$

$$\begin{aligned} \hat{H} &= \hat{H}_0 + \frac{e}{m_e} \vec{p} \cdot \vec{A} + \frac{e^2}{2m_e} \vec{A}^2 \\ &= \hat{H}_0 + \frac{e}{2m_e} \vec{p} \cdot (\vec{B} \times \vec{r}) + \frac{e^2}{8m_e} (\vec{B} \times \vec{r})^2 \quad \vec{p} \cdot (\vec{B} \times \vec{r}) = \vec{B} \cdot (\vec{r} \times \vec{p}) = \vec{B} \cdot \vec{L} \\ &= \hat{H}_0 + \frac{e}{2m_e} \vec{B} \cdot \vec{L} + \frac{e^2}{8m_e} (\vec{B} \times \vec{r})^2 \\ &\approx \hat{H}_0 + \frac{e}{2m_e} \vec{B} \cdot \vec{L} = \hat{H}_0 - \hat{\mu}_L \cdot \vec{B} \quad \text{Bohr 磁子 } \hat{\mu}_L = -\frac{e\hbar}{2m_e} \end{aligned}$$

第三项
第二项 $\sim \frac{eB^2 r^2}{\hbar} \ll 1$

自旋与磁场的相互作用: $\hat{H}_S = -\hat{\mu}_S \cdot \vec{B}$ $\hat{\mu}_S = -\frac{e}{m_e} \vec{S}$

$$g \text{ 因子: } \begin{cases} \hat{\mu}_L = -g_L \frac{m_B}{\hbar} \vec{L}, & g_L = 2 \\ \hat{\mu}_S = -g_S \frac{m_B}{\hbar} \vec{S}, & g_S = 1 \end{cases}$$

3. Landau 能级

(1) Landau 规范 具有平移不变性.

粒子在沿 z 方向的均匀磁场 $\vec{B} = B \hat{z}$ 中运动

选取规范: $\vec{A} = (0, Bx, 0)$

$$\hat{H} = \frac{1}{2m} \vec{p}^2 + \frac{1}{2m} (p_y - qBx)^2 + \frac{1}{2m} p_z^2$$

$[p_y, \hat{H}] = 0$, $[p_z, \hat{H}] = 0 \rightarrow$ 沿 y, z 方向有平移对称性.

$$\psi(x, y, z) = \phi(x) e^{\frac{i}{\hbar} (p_y y + p_z z)}$$

Schrodinger 方程: $\hat{H}\psi = E\psi$

$$\frac{1}{2m} \left[p_x^2 + (p_y - qBx)^2 + p_z^2 \right] \psi = E\psi$$

$$= \frac{1}{2m} \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \psi + \frac{1}{2m} \left[(p_y - qBx)^2 + p_z^2 \right] \psi = E\psi$$

$$= -\frac{\hbar^2}{2m} \frac{d^2 \phi(x)}{dx^2} + \frac{q^2 B^2}{2m} \left(x - \frac{p_y}{qB} \right)^2 \phi(x) = \left(E - \frac{p_z^2}{2m} \right) \phi(x) = \varepsilon \phi(x)$$

$$\omega = \frac{qB}{m}, \quad x_0 = \frac{p_y}{qB}$$

$$\omega = -\frac{\hbar^2}{2m} \frac{d^2 \phi(x)}{dx^2} + \frac{1}{2} m \omega^2 (x - x_0)^2 \phi(x) = \left(\epsilon - \frac{p_z^2}{2m} \right) \phi(x)$$

$$\Rightarrow \epsilon_n = \left(n + \frac{1}{2} \right) \hbar \omega + \frac{p_z^2}{2m} = \left(n + \frac{1}{2} \right) \frac{qB \hbar}{m} + \frac{p_z^2}{2m}, \text{ 为 Landau 能级.}$$

相应本征函数:

$$\psi_n(x, y, z) = \frac{1}{\sqrt{2\pi}} \left(\frac{b}{\sqrt{\pi}} \right)^{1/2} \frac{H_n \left[\frac{b}{\sqrt{\pi}} (x - x_0) \right]}{\sqrt{2^n n!}} \exp \left[-\frac{b^2}{2} (x - x_0)^2 \right] e^{\frac{i}{\hbar} (p_y y + p_z z)}$$

简并度: $D = \frac{\phi}{\phi_0}$, $\phi_0 = \frac{2\pi \hbar^2}{e}$, 为磁通量子

2) 对称规范

考虑均匀磁场中进行二维运动的粒子

$$\hat{H} = \frac{1}{2m} [(\hat{p}_x - q\hat{A}_x)^2 + (\hat{p}_y - q\hat{A}_y)^2] = \frac{1}{2m} (\hat{\pi}_x^2 + \hat{\pi}_y^2)$$

$$[\hat{\pi}_x, \hat{\pi}_y] = [\hat{p}_x - q\hat{A}_x, \hat{p}_y - q\hat{A}_y]$$

$$= q(\hat{p}_y \hat{A}_x - \hat{p}_x \hat{A}_y) = i\hbar q \left(\frac{\partial}{\partial x} \hat{A}_y - \frac{\partial}{\partial y} \hat{A}_x \right)$$

$$= i\hbar q B_z$$

$$\text{定义升降算符 } \hat{a} = \frac{1}{\sqrt{2\hbar q B}} (\hat{\pi}_x + i\hat{\pi}_y),$$

$$\text{消灭算符 } \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar q B}} (\hat{\pi}_x - i\hat{\pi}_y)$$

$$\hookrightarrow \hat{\pi}_x = \sqrt{\frac{\hbar q B}{2}} (\hat{a} + \hat{a}^\dagger), \quad \hat{\pi}_y = -i \sqrt{\frac{\hbar q B}{2}} (\hat{a} - \hat{a}^\dagger)$$

$$\hat{a}^\dagger \hat{a} = \frac{1}{2\hbar q B} [\hat{\pi}_x^2 + \hat{\pi}_y^2 + i(\hat{\pi}_x \hat{\pi}_y - \hat{\pi}_y \hat{\pi}_x)]$$

$$= \frac{1}{2\hbar q B} (\hat{\pi}_x^2 + \hat{\pi}_y^2 - \hbar q B) = \frac{m}{\hbar q B} \cdot \frac{1}{2m} (\hat{\pi}_x^2 + \hat{\pi}_y^2) - \frac{1}{2}$$

$$\hat{H} = \hbar \omega (\hat{a}^\dagger \hat{a} + \frac{1}{2}), \quad \omega = \frac{qB}{m}$$

$$\epsilon_n = \hbar \omega \left(n + \frac{1}{2} \right), \text{ Landau 能级.}$$

求对应本征态:

$$\hat{A} = \frac{B}{2} (-\hat{y}, \hat{x}, 0)$$

$$\hat{H} = \frac{1}{2m} [(\hat{p}_x - q\hat{A}_x)^2 + (\hat{p}_y - q\hat{A}_y)^2]$$

$$= \frac{1}{2m} [(\hat{p}_x + \frac{1}{2} qB\hat{y})^2 + (\hat{p}_y - \frac{1}{2} qB\hat{x})^2]$$

$$= \frac{1}{2m} (\hat{p}_x^2 + qB \hat{p}_x \hat{y} + \frac{1}{4} q^2 B^2 \hat{y}^2 + \hat{p}_y^2 - qB \hat{p}_y \hat{x} + \frac{1}{4} q^2 B^2 \hat{x}^2)$$

$$= \frac{1}{2m} (\hat{p}_x^2 + \hat{p}_y^2) - \frac{qB}{2m} \hat{L}_z + \frac{q^2 B^2}{8m} (\hat{x}^2 + \hat{y}^2)$$

有绕z轴的旋转对称性.

$[\hat{L}_z, \hat{H}] = 0 \rightarrow$ 能量与磁态可表为 $|n, m\rangle$

$$\hat{H} |n, m\rangle = E_n |n, m\rangle, \quad \hat{L}_z |n, m\rangle = m\hbar |n, m\rangle$$

$$\hat{z} \times \hat{r}' = \hat{p} + q\hat{B}$$

$$1) [\hat{\lambda}_x', \hat{\lambda}_y'] = -i\hbar qB$$

$$2) [\hat{\lambda}_x, \hat{\lambda}_x'] = [\hat{\lambda}_y, \hat{\lambda}_y'] = [\hat{\lambda}_x, \hat{\lambda}_y'] = [\hat{\lambda}_y, \hat{\lambda}_x'] = 0$$

$$3) \hat{L}_z = \frac{1}{2qB} (\hat{\lambda}_x'^2 + \hat{\lambda}_y'^2 - \hat{\lambda}_x^2 - \hat{\lambda}_y^2)$$

$$\hat{\lambda}_x'^2 - \hat{\lambda}_x^2 + \hat{\lambda}_y'^2 - \hat{\lambda}_y^2$$

$$= 4q(\hat{p}_x \hat{A}_x + \hat{p}_y \hat{A}_y) = 2qB(\hat{p}_x (-\hat{y}) + \hat{p}_y x)$$

$$\hat{z} \times \hat{b} = \frac{1}{(2\hbar qB)} (\hat{\lambda}_x' + i\hat{\lambda}_y'), \quad \hat{b}^\dagger = \frac{1}{(2\hbar qB)} (\hat{\lambda}_x' - i\hat{\lambda}_y')$$

$$\hat{\lambda}_x' = \sqrt{\frac{\hbar qB}{2}} (\hat{b} + \hat{b}^\dagger), \quad \hat{\lambda}_y' = -\sqrt{\frac{\hbar qB}{2}} i(\hat{b} - \hat{b}^\dagger)$$

$$[\hat{b}, \hat{b}^\dagger] = \frac{1}{2\hbar qB} (2i\hbar [\hat{\lambda}_x', \hat{\lambda}_y']) = 1$$

$$[\hat{a}, \hat{b}] = \frac{1}{2\hbar qB} \{ [\hat{\lambda}_x, \hat{\lambda}_x'] + i([\hat{\lambda}_y, \hat{\lambda}_x'] + [\hat{\lambda}_x', \hat{\lambda}_y']) - [\hat{\lambda}_y, \hat{\lambda}_y'] \} = 0$$

$$[\hat{a}^\dagger, \hat{b}^\dagger] = 0$$

$$\begin{aligned} \hat{L}_z &= \frac{\hbar}{4} [(\hat{b} + \hat{b}^\dagger)^2 - (\hat{b} - \hat{b}^\dagger)^2 - (\hat{a} + \hat{a}^\dagger)^2 + (\hat{a} - \hat{a}^\dagger)^2] \\ &= \hbar (\hat{b}^\dagger \hat{b} - \hat{a}^\dagger \hat{a}) \end{aligned}$$

$$\hat{L}_z (\hat{b}^\dagger |n, m\rangle) = (m+1)\hbar |n, m\rangle, \quad \hat{L}_z (\hat{a}^\dagger |n, m\rangle) = (m-1)\hbar |n, m\rangle$$

$$\hookrightarrow |n, m\rangle = \frac{\hat{a}^\dagger}{\sqrt{n!}} \frac{(\hat{b}^\dagger)^{n+m}}{(n+m)!} |0, 0\rangle$$

求磁态的具像方程: $\omega = x + iy$

$$\begin{cases} \hat{a} = -i \sqrt{\frac{\hbar}{2qB}} (2 \frac{\partial}{\partial \omega^*} + \frac{qB}{2\hbar} \omega) \\ \hat{a}^\dagger = -i \sqrt{\frac{\hbar}{2qB}} (2 \frac{\partial}{\partial \omega} - \frac{qB}{2\hbar} \omega) \end{cases}$$

$$\begin{cases} \hat{b} = \sqrt{\frac{\hbar}{2qB}} (2 \frac{\partial}{\partial \omega} + \frac{qB}{2\hbar} \omega) \\ \hat{b}^\dagger = \sqrt{\frac{\hbar}{2qB}} (-2 \frac{\partial}{\partial \omega^*} + \frac{qB}{2\hbar} \omega) \end{cases}$$

$$\hat{a} = \frac{1}{\sqrt{2q\hbar B}} (\hat{\lambda}_x' + i\hat{\lambda}_y')$$

$$= \frac{1}{\sqrt{2q\hbar B}} (\hat{p}_x + i\hat{p}_y) + q(\hat{A}_x + i\hat{A}_y)$$

$$= -i \sqrt{\frac{\hbar}{2qB}} (2 \frac{\partial}{\partial \omega} + \frac{qB}{2\hbar} \omega) + \frac{qB}{2\hbar} (x + iy)$$

$$= -i \sqrt{\frac{\hbar}{2qB}} (2 \frac{\partial}{\partial \omega^*} + \frac{qB}{2\hbar} \omega)$$

对于真空态 $|0,0\rangle$

$$\hat{a}|0,0\rangle = 0, \hat{b}|0,0\rangle = 0$$

$$\langle w, w^* | \hat{a} | 0, 0 \rangle = -i \sqrt{\frac{\hbar}{2q\hbar^3}} \left(2 \frac{\partial}{\partial w^*} + \frac{q\hbar}{2\hbar} w \right) \langle w, w^* | 0, 0 \rangle = 0$$

$$\text{解得 } \psi_{0,0}(w, w^*) = \langle w, w^* | 0, 0 \rangle = f(w) e^{-\frac{q\hbar}{4\hbar} |w|^2}$$

$$\langle w, w^* | \hat{b} | 0, 0 \rangle = \sqrt{\frac{\hbar}{2q\hbar^3}} \left(2 \frac{\partial}{\partial w} + \frac{q\hbar}{2\hbar} w^* \right) \langle w, w^* | 0, 0 \rangle = 0$$

$$\text{解得 } f(w) = C$$

$$\hookrightarrow \psi_{0,0}(w, w^*) = C_{0,0} e^{-\frac{q\hbar}{4\hbar} |w|^2}$$

$$\psi_{n,m}(w, w^*) = \langle w, w^* | n, m \rangle$$

$$= \frac{1}{n!(n+m)!} \left[-i \sqrt{\frac{\hbar}{2q\hbar^3}} \left(2 \frac{\partial}{\partial w} - \frac{q\hbar}{2\hbar} w \right) \right]^n \left[\sqrt{\frac{\hbar}{2q\hbar^3}} \left(-2 \frac{\partial}{\partial w} + \frac{q\hbar}{2\hbar} w^* \right) \right]^{m,n} \psi_{0,0}(w, w^*)$$

$$= C_{n,m} \left(\frac{\partial}{\partial w} - \frac{q\hbar}{4\hbar} w^* \right)^n (w^{n+m} e^{-\frac{q\hbar}{4\hbar} |w|^2})$$

AB 效应

引入磁场前后, 具有相同能量的本征函数 $\phi(\vec{r})$, $\psi(\vec{r})$

$$\psi(\vec{r}) = \phi(\vec{r}) e^{\frac{iq}{\hbar} \int_0^{\vec{r}} \vec{A} \cdot d\vec{r}'}$$

$$\text{证: 已知 } \left[\frac{1}{2m} (\vec{p} - q\vec{A})^2 + V(\vec{r}) \right] \psi(\vec{r}) = E \psi(\vec{r})$$

$$= \left[\frac{1}{2m} (\vec{p} - q\vec{A})^2 + V(\vec{r}) \right] \phi(\vec{r}) e^{\frac{iq}{\hbar} \int_0^{\vec{r}} \vec{A} \cdot d\vec{r}'} \quad [\vec{p}, f(\vec{r})]$$

$$= e^{\frac{iq}{\hbar} \int_0^{\vec{r}} \vec{A}(\vec{r}') \cdot d\vec{r}'} \left[\frac{1}{2m} (\vec{p} - q\vec{A} + q\vec{A})^2 + V(\vec{r}) \right] \phi(\vec{r})$$

$$= E \phi(\vec{r})$$

$$\psi(\vec{r}) = \psi^{(1)}(\vec{r}) + \psi^{(2)}(\vec{r})$$

$$= e^{\frac{iq}{\hbar} \int_0^{\vec{r}} \vec{A} \cdot d\vec{r}'} (\phi_0^{(1)} + \phi_0^{(2)}) e^{\frac{iq}{\hbar} \phi \vec{A} \cdot d\vec{r}'} \quad \oint \vec{A} \cdot d\vec{r} = \int_V \vec{B} \cdot d\vec{S} = \Phi$$

Φ 是一个规范不变量。

磁通量子化:

$$\text{相位差 } e^{\frac{iq}{\hbar} \Phi} \text{ 单值 } \frac{q\Phi}{\hbar} = 2n\pi$$

$$\hookrightarrow \Phi_0 = \frac{2\pi\hbar}{e}$$

作业:

1. $r > a$, $\psi = 0$

$$r < \frac{a}{2} , \quad \frac{d^2 u}{dr^2} + k^2 u(r) = 0$$

$$\lim_{r \rightarrow 0} r^2 V(r) = 0 \rightarrow \cos r \text{ 不满足}, u(r) = A \sin kr$$

$$\frac{a}{2} < r < a , \quad \frac{d^2 u}{dr^2} - k^2 u = 0 , \quad k = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

$$u(r) = B e^{kr} + C e^{-kr}$$

$$\begin{cases} A \sin(k \frac{a}{2}) = B e^{k \frac{a}{2}} + C e^{-k \frac{a}{2}} \\ k A \cos(k \frac{a}{2}) = k B e^{k \frac{a}{2}} - k C e^{-k \frac{a}{2}} \\ B e^{ka} + C e^{-ka} = 0 \end{cases}$$

2. (1)

$$2) \quad \psi = R Y$$

$$R = C e^{-\frac{r}{a}}$$

概率密度 $\rho(r) = R^2 r^2$

$$\frac{d\rho(r)}{dr} = 0 \rightarrow r = a$$

(3) $R(r)$

3. 时间演化算符 $\hat{U} = \begin{pmatrix} e^{-iZ_1 \frac{\Delta\tau}{\hbar}} & 0 \\ 0 & e^{-iZ_2 \frac{\Delta\tau}{\hbar}} \end{pmatrix}$

$$\begin{cases} -Z_1 \frac{\Delta\tau}{\hbar} = 2\pi n \\ -Z_2 \frac{\Delta\tau}{\hbar} = 2\pi m + \varphi \end{cases}$$

$$\hat{H} = \begin{pmatrix} Z_1 & 0 \\ 0 & Z_2 \end{pmatrix} = \frac{\lambda \hbar}{\Delta\tau} (N + \frac{\varphi}{2\lambda}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \text{const.}$$

(2) $\hat{Q} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$\hat{Q}^\dagger \hat{U}_{\text{time}} \hat{Q} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{cases} Z_1 \frac{\Delta\tau}{\hbar} = 2\pi n \\ Z_2 \frac{\Delta\tau}{\hbar} = (2m+1)\pi \end{cases}$$

$$\hat{H}' = \frac{\lambda \hbar}{\Delta\tau} (N + \frac{1}{2}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\hat{H} = \hat{Q} \hat{H}' \hat{Q}^\dagger = \frac{\lambda \hbar}{\Delta\tau} (N+1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

4. (1) 选取基矢: $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$ (直接)

$$\hat{H} = J \hat{S}_1^z \hat{S}_2^z + (J_0 - J) \hat{S}_1^x \hat{S}_2^x$$

$$[\hat{S}_1^z \hat{S}_2^z, \hat{S}_1^x \hat{S}_2^x]$$

$$= \hat{S}_1^z \hat{S}_2^z \hat{S}_1^x \hat{S}_2^x - \hat{S}_1^x \hat{S}_2^x \hat{S}_1^z \hat{S}_2^z$$

$$= \hat{S}_1^z \hat{S}_1^x \hat{S}_2^z \hat{S}_2^x - \hat{S}_1^x \hat{S}_1^z \hat{S}_2^x \hat{S}_2^z$$

不同角动之间可交换

反对易。

$$=$$

$$= 0$$

$$[\hat{S}_1^z \hat{S}_2^z, \hat{S}_1^x \hat{S}_2^x]$$

$\Rightarrow \hat{H}$ 可用 $\hat{S}_1^z \hat{S}_2^z, \hat{S}_1^x \hat{S}_2^x$ 的共同本征态表示。

5. (1) 归一化 $\rho = \frac{1}{\sqrt{6}}$

$$\langle \hat{L}_z \rangle = \frac{1}{6} \times 0 + \frac{1}{6} \hbar + \frac{2}{3} (-\hbar) = -\frac{1}{3} \hbar$$

$$\langle \hat{S}_z \rangle = -\frac{1}{6} \hbar$$

(2) $[\hat{L}_x, \hat{L}^2] = 0 \rightarrow$ 可按 L_x 来划分子空间.

① $l=0$ $|100\rangle$ 为 L_x 的本征态, 本征值为 0

$$\langle 100, \uparrow | \psi \rangle = \frac{1}{\sqrt{6}}, \quad \langle 100, \downarrow | \psi \rangle = 0$$

② $l=1$ 选取 $\{|211\rangle, |210\rangle, |21-1\rangle\}$ 为基矢

$$\hat{L}_x = \frac{\sqrt{2}}{2} \hbar \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \rightarrow L_x = \pm \hbar, 0$$

$$\langle 2, 1, L_x = -\hbar, \uparrow | \psi \rangle = \frac{1}{2\sqrt{6}}, \quad \langle 2, 1, L_x = -\hbar, \downarrow | \psi \rangle = \frac{1}{\sqrt{6}}$$

$$\langle 2, 1, L_x = 0, \uparrow | \psi \rangle = \frac{1}{2\sqrt{3}}, \quad \langle 2, 1, L_x = 0, \downarrow | \psi \rangle = \frac{1}{\sqrt{3}}$$

$$\langle 2, 1, L_x = \hbar, \uparrow | \psi \rangle = \frac{1}{2\sqrt{6}}, \quad \langle 2, 1, L_x = \hbar, \downarrow | \psi \rangle = \frac{1}{\sqrt{6}}$$

$$P(L_x = -\hbar) = \frac{5}{24}$$

$$P(L_x = 0) = \frac{7}{12}$$

$$P(L_x = \hbar) = \frac{5}{24}$$

(3) $\langle \hat{L}^2 \rangle = \hbar^2 \rightarrow \langle \hat{L}_z \rangle = \pm \hbar$

$$\frac{1}{\sqrt{3}} \begin{pmatrix} \psi_{211} \\ \sqrt{2} \psi_{21-1} \end{pmatrix}$$

6. 选取规范 $\hat{A} = (0, B\hat{x}, 0)$

$$\hat{p}_y' = \hat{p}_y + eB\hat{x}$$

$$\hat{H}' = \begin{pmatrix} 0 & \hat{p}_x - i\hat{p}_y' - ieB\hat{x} \\ \hat{p}_x + i\hat{p}_y' + ieB\hat{x} & 0 \end{pmatrix}$$

考虑 $\hat{H}|\psi\rangle = E|\psi\rangle$, $\hat{H} \begin{pmatrix} \psi_x(x, y) \\ \psi_y(x, y) \end{pmatrix} = E \begin{pmatrix} \psi_x(x, y) \\ \psi_y(x, y) \end{pmatrix}$

{

$$\Rightarrow v_f^2 [\hat{p}_x^2 + \hat{p}_y'^2 - eB\hbar + 2eB\hat{x}\hat{p}_y' + e^2B^2\hat{x}^2] \psi_y(x, y) = E^2 \psi_y(x, y)$$

平移对称性: $\psi_y(x, y) = \mu(x) e^{-\frac{i}{\hbar} p_y y}$

$$[v_f^2 \hat{p}_x^2 + v_f^2 e^2 B^2 (x + \frac{p_y}{eB})^2] \mu(x) = (E^2 + eB\hbar v_f^2) \mu(x)$$

$$m = \frac{1}{2v_f^2}, \quad \omega = 2v_f^2 eB\hbar$$

$$E^2 + eB\hbar v_f^2 = (n + \frac{1}{2}) \omega$$

$$E = \pm v_f \sqrt{2eB\hbar n}$$

$$\hat{\pi}_x = \hat{p}_x + e\hat{A}_x, \quad \hat{\pi}_y = \hat{p}_y + e\hat{A}_y$$

$$\begin{cases} \hat{a} = \frac{1}{\sqrt{2\hbar eB}} (\hat{\pi}_x + i\hat{\pi}_y) \\ \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar eB}} (\hat{\pi}_x - i\hat{\pi}_y) \end{cases}$$

$$\hat{H} = v_f \sqrt{2\hbar eB} \begin{pmatrix} 0 & \hat{a} \\ \hat{a}^\dagger & 0 \end{pmatrix} = \hbar\omega \begin{pmatrix} 0 & \hat{a} \\ \hat{a}^\dagger & 0 \end{pmatrix}$$

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

$$\hat{H}^2|\psi\rangle = E^2|\psi\rangle \quad \hat{H}^2 = \hbar^2\omega^2 \begin{pmatrix} \hat{a}\hat{a}^\dagger & 0 \\ 0 & \hat{a}^\dagger\hat{a} \end{pmatrix}$$

$$\begin{cases} \hat{H}^2|\psi_1\rangle = \hbar^2\omega^2 \hat{a}\hat{a}^\dagger|\psi_1\rangle = \hbar^2\omega^2 (\hat{a}^\dagger\hat{a} + 1)|\psi_1\rangle = E^2|\psi_1\rangle \\ \hat{H}^2|\psi_2\rangle = \hbar^2\omega^2 \hat{a}^\dagger\hat{a}|\psi_2\rangle = E^2|\psi_2\rangle \end{cases}$$

$$\Rightarrow \begin{cases} |\psi_1\rangle = |n-1\rangle \\ |\psi_2\rangle = |n\rangle \end{cases}$$